AdaBoost: Run Weights, Vote Weights, Exponential Loss, and AdaBoost.R2

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia Sources: `Lectures/L12_student_trees_bagging_boosting_worked_deck.pdf` (Part 4, slides 65–81), `Lectures/L12e_student_adaboost_classification_notes.pdf` (12-page notes), `Lectures/L12f_student_adaboost_regression_notes.pdf` (13-page notes) · Previous: Gradient boosting · Next: XGBoost and comparison

Lecture 12 files: L12a Trees → L12b Bagging & random forests → L12c Gradient boosting → **L12d AdaBoost** → L12e XGBoost & comparison

> **How to use this file.** Gradient boosting (L12c) passed mistakes to the next stump as **residuals**. AdaBoost keeps the original labels and passes mistakes on as **weights**: a run the stumps so far get wrong **counts more** in the next round. The final model is a **weighted vote** of the stumps. Four numbers drive everything: run weights w_i, weighted error ε, vote weight $\alpha = \frac{1}{2}\ln\frac{1-\varepsilon}{\varepsilon}$, normaliser Z. We derive α and the weight update from one loss, the **exponential loss** e^{-yF}, and show AdaBoost is gradient boosting for that loss. Then **AdaBoost.R2** extends the idea to regression with a **weighted median**. Part A is maths step by step; Part B walks slides 65–81; Part B′ covers the two notes; Part C is practice.

Datasets. The twelve runs of L12a, now with **labels ±1**: $y = +1$ for D (runs 1–6), -1 for C (runs 7–12); new runs **A** (0.05, 32, no), **B** (0.2, 64, yes). For AdaBoost.R2: eight points (minutes per epoch vs model size):

i	1	2	3	4	5	6	7	8
x	1	2	3	4	5	6	7	8
y	1.5	1.5	4	4	5.5	7	7	8.5

What this file is really about:

$$w_i = \tfrac{1}{N} \;\to\; \text{stump with smallest } \varepsilon = \sum_{\text{mistakes}} w_i \;\to\; \alpha = \tfrac{1}{2}\ln\tfrac{1-\varepsilon}{\varepsilon} \;\to\; w_i \leftarrow \frac{w_i e^{-\alpha y_i h(x_i)}}{Z} \;\to\; \text{repeat} \;\to$$

Step-up path

You need. Stumps and weighted averages.

1. Weight the points, and find the stump with the lowest weighted error.
2. Give that stump a vote based on its error.
3. Raise the weight of its mistakes, and repeat.
4. Derive the vote by minimising Z.
5. See AdaBoost as boosting on the exponential loss.

You can stop when you can explain why the stump just used scores exactly 50% on the new weights.

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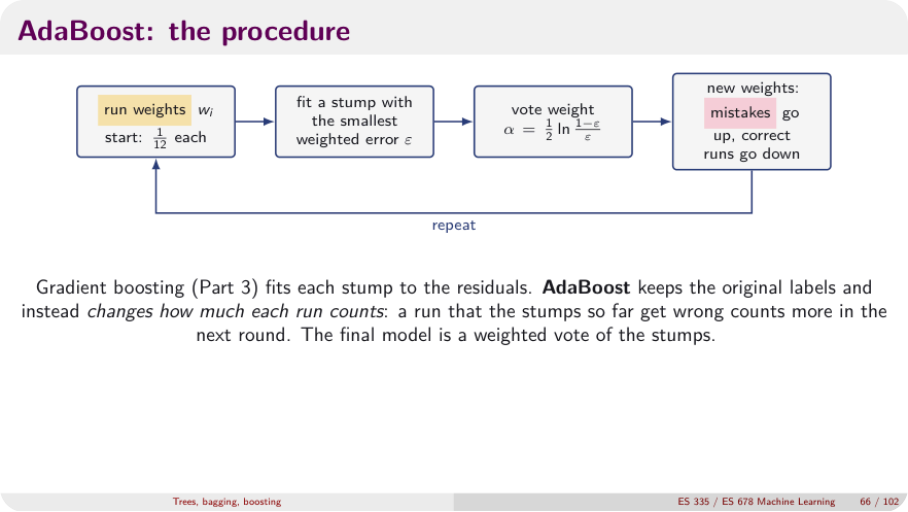
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Part A Maths From the Ground Up

A1 The procedure and the definitions



Slide 66 — AdaBoost changes **how much each run counts**, not what the stumps are fitted to.

Definitions (slide 67):

- **Labels** $y_i = \pm 1$. A stump gives $h(x) = \pm 1$; each leaf predicts the class with the larger **total weight** in that leaf.
- **Run weights** $w_i \geq 0$, $\sum w_i = 1$; start $w_i = \frac{1}{12} = 0.0833$. Think: "how much this run counts when we count mistakes".
- **Weighted error** $\varepsilon = \sum_{i:\, h(x_i) \neq y_i} w_i$ — total weight of the mistakes.
- **Vote weight** $\alpha = \frac{1}{2} \ln \frac{1-\varepsilon}{\varepsilon}$ — large when ε small; 0 at ε = ½; negative if ε > ½.
- **Update** $w_i \leftarrow w_i e^{-\alpha y_i h(x_i)} / Z$: since $y_i h(x_i) = +1$ if right and −1 if wrong, a correct run is multiplied by $e^{-\alpha}$ (shrinks), a mistake by $e^{+\alpha}$ (grows); Z makes them add to 1.
- **Model** $F(x) = \sum_t \alpha_t h_t(x)$; predict D if $F > 0$.

Rounds = stumps. One round = pick one stump, measure ε, compute α, re-weight. After T rounds there are T stumps. T is chosen on validation data.

A2 Round 1: choose the stump, compute ε and α

Step 1 — Equal weights ⇒ ε = (number of mistakes)/12.

question	yes-leaf says	no-leaf says	mistakes	ε
rate ≤ 0.065	C	D	2	**0.1667**
rate ≤ 0.02	C	D	3	0.2500
rate ≤ 0.0055	C	D	4	0.3333
rate ≤ 0.2	C	D	4	0.3333
batch ≤ 48	D	C	4	0.3333
warm-up = no	D	C	4	0.3333
batch ≤ 96	D	C	5	0.4167
batch ≤ 192	D	C	5	0.4167

Your turn 1 (warm-up = no): yes-leaf {2, 4, 5, 6, 7, 11} has 4 D, 2 C ⇒ says D; no-leaf {1, 3, 8, 9, 10, 12} has 2 D, 4 C ⇒ says C. Wrong: 7, 11, 1, 3 ⇒ ε = 4/12.

Step 2 — Keep the smallest ε. rate ≤ 0.065: yes (runs 5–12: 6 C, 2 D) ⇒ C; no (runs 1–4, all D) ⇒ D. Mistakes: **runs 5 and 6**.

$$\varepsilon_1 = w_5 + w_6 = \tfrac{2}{12} = 0.1667, \qquad \alpha_1 = \tfrac{1}{2}\ln\frac{0.8333}{0.1667} = \tfrac{1}{2}\ln 5 = 0.805.$$

Step 3 — How α behaves (slides 69–70):

ε	0.1	0.3	0.5	0.7
α	½ ln 9 = 1.099	½ ln(7/3) = 0.424	½ ln 1 = 0	½ ln(3/7) = −0.424

ε = 0.1 gets the loudest voice; ε = 0.5 (coin toss) gets **no** vote; ε = 0.7 gets a **negative** vote: AdaBoost uses it by **flipping its answer**. If ε → 0, α → ∞ (a perfect stump would take over), so implementations cap α. *Your turn 2:* ε = 0.25 ⇒ α = ½ ln 3 = 0.549 (quieter than 0.805).

A3 Round 1: update the weights

Step 1 — Multipliers. $e^{-\alpha_1} = 1/\sqrt{5} = 0.447$ (right), $e^{+\alpha_1} = \sqrt{5} = 2.236$ (wrong).

Step 2 — Multiply, add up, divide.

runs	right?	w before	multiplier	w × mult.	w after (÷ Z)
1–4, 7–12 (ten runs)	yes	0.0833	0.447	0.0373	**0.0500**
5, 6	**no**	0.0833	2.236	0.1863	**0.2500**

$Z_1 = 10(0.0373) + 2(0.1863) = 0.7454$. Check: $10 \times 0.05 + 2 \times 0.25 = 1$ ✓. *Your turn 3:* run 1: $0.0373/0.7454 = 0.050$; run 5: $0.1863/0.7454 = 0.250$.

Step 3 — What just happened. Runs 5, 6 jumped from 0.083 to **0.250** each; together they hold **half of all the weight**.

A4 Why this α: minimise Z

Step 1 — Total weight after multiplying (before normalising), with weights summing to 1:

$$Z(\alpha) = (1-\varepsilon)e^{-\alpha} + \varepsilon e^{\alpha}.$$

Step 2 — Choose α to make it smallest.

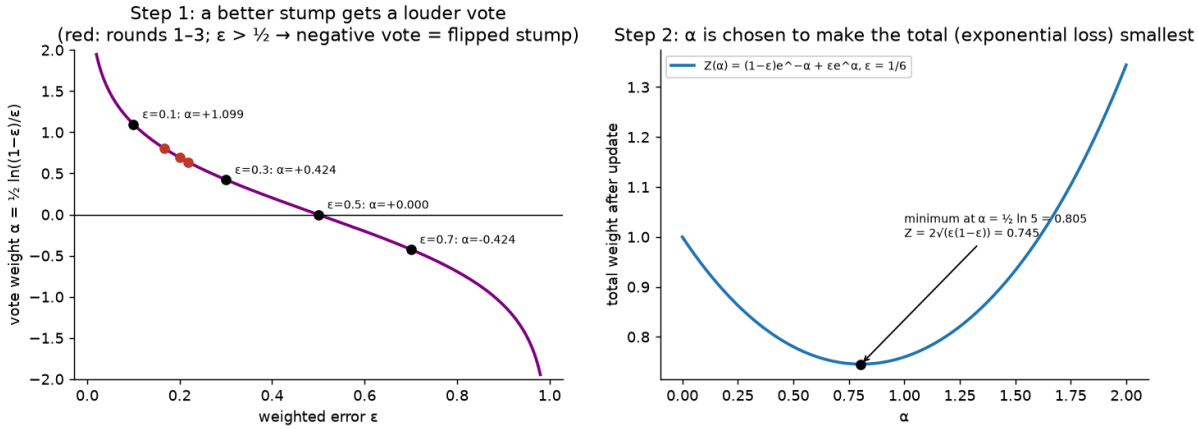
$$\frac{dZ}{d\alpha} = -(1-\varepsilon)e^{-\alpha} + \varepsilon e^{\alpha} = 0 \;\Rightarrow\; e^{2\alpha} = \frac{1-\varepsilon}{\varepsilon} \;\Rightarrow\; \alpha = \tfrac{1}{2}\ln\frac{1-\varepsilon}{\varepsilon}.$$

(Second derivative $(1-\varepsilon)e^{-\alpha} + \varepsilon e^{\alpha} > 0$: a minimum.)

Step 3 — The minimum value. With $e^{\alpha} = \sqrt{(1-\varepsilon)/\varepsilon}$:

$$Z = \varepsilon\sqrt{\tfrac{1-\varepsilon}{\varepsilon}} + (1-\varepsilon)\sqrt{\tfrac{\varepsilon}{1-\varepsilon}} = 2\sqrt{\varepsilon(1-\varepsilon)}.$$

For ε = 1/6: $2\sqrt{5/36} = 0.7454$ — exactly Z_1. Why minimise Z? Because Z is the factor by which the **exponential loss** shrinks this round (A8–A9).



Step 1: the vote weight curve. Step 2: α is exactly the value that makes the total weight (the exponential loss) smallest.

A5 The key consequence: the old stump becomes a coin toss

Step 1 — Mistakes' share after the update:

$$\frac{\varepsilon e^{\alpha}}{Z} = \frac{\varepsilon\sqrt{(1-\varepsilon)/\varepsilon}}{2\sqrt{\varepsilon(1-\varepsilon)}} = \frac{1}{2}.$$

The correct runs carry the other ½.

Step 2 — So the stump just used has weighted error exactly ½ on the new weights. Reusing it now would give α = 0. The next stump **must find something different** — here, something that handles runs 5 and 6. *That is the whole mechanism.*

Step 3 — Check: after round 1, rate ≤ 0.065 errs on runs 5, 6: $0.25 + 0.25 = 0.5 \Rightarrow$ α = 0 ✓.

A6 Round 2 and the tie rule

Step 1 — Weights: runs 5, 6 at 0.25; the rest at 0.05. Leaves now predict the class with the larger **weight**.

question (round 2)	yes-leaf	no-leaf	ε
rate ≤ 0.0055	C	D	0.2000
batch ≤ 48	D	C	**0.2000** (chosen)
warm-up = no	D	C	0.2000
batch ≤ 96	D	C	0.2500
batch ≤ 192	D	C	0.2500
rate ≤ 0.02	D	D	0.3000

Step 2 — Work batch ≤ 48 by hand (*your turn 4*). Yes-leaf {2, 5, 6, 10}: D weight $0.05 + 0.25 + 0.25 = 0.55$, C weight 0.05 ⇒ says D. No-leaf {1, 3, 4, 7, 8, 9, 11, 12}: D weight 0.15, C weight 0.25 ⇒ says C. Mistakes: run 10 and runs 1, 3, 4 ⇒

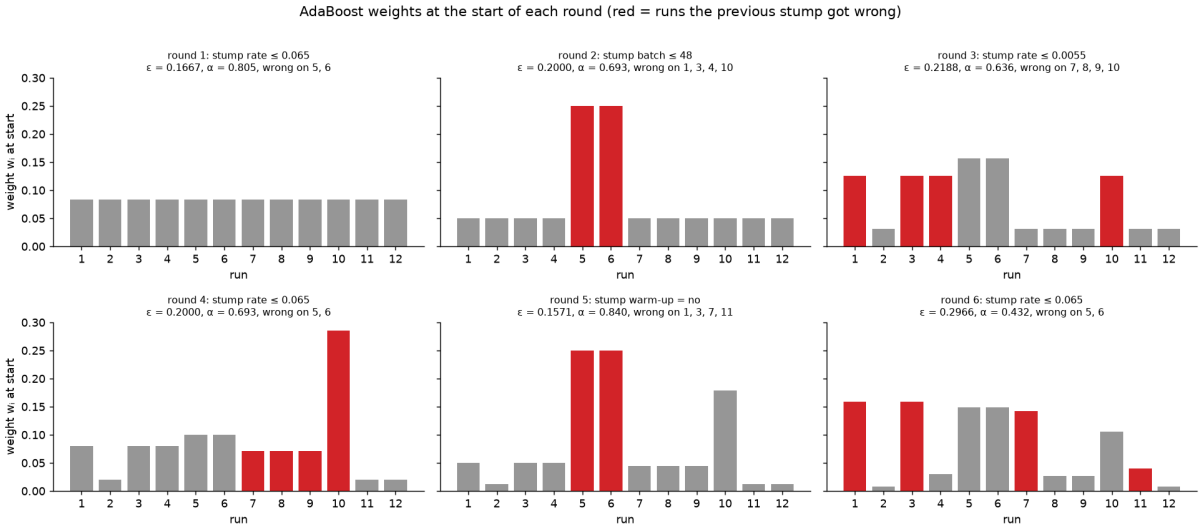
$$\varepsilon_2 = 4 \times 0.05 = 0.200 \;\; (\text{ordinary error } 4/12 = 0.333), \qquad \alpha_2 = \tfrac{1}{2} \ln \tfrac{0.8}{0.2} = \tfrac{1}{2} \ln 4 = 0.693.$$

The weighted error is smaller than the ordinary one because its mistakes are runs that already count little.

Step 3 — Tie rule. Three questions tie at 0.2. Rule (deck and notes): prefer the feature **asked fewest times so far**, then the order rate, batch, warm-up ⇒ **batch**.

Step 4 — Update. Multipliers $e^{\mp 0.693} = 0.5,\ 2.0$; $Z_2 = 0.8$:

runs	right?	before	after
1, 3, 4, 10	no	0.05	**0.125**
5, 6	yes	0.25	**0.156**
2, 7, 8, 9, 11, 12	yes	0.05	**0.031**

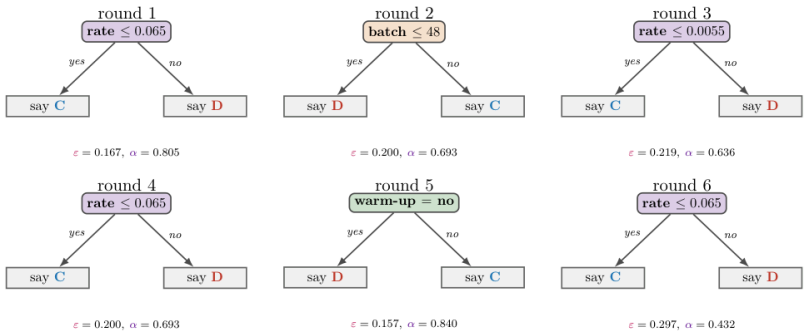


Weight flows to whatever the last stump got wrong: 5, 6 after round 1; 1, 3, 4, 10 after round 2; 7, 8, 9, 10 after round 3; and so on.

A7 All rounds; the final weighted vote

round	stump	yes, no say	ε	α	stump's mistakes	committee's mistakes	mean e^{-yF}
1	rate ≤ 0.065	C, D	0.1667	0.805	5, 6	5, 6	0.745
2	batch ≤ 48	D, C	0.2000	0.693	1, 3, 4, 10	5, 6	0.596
3	rate ≤ 0.0055	C, D	0.2188	0.636	7, 8, 9, 10	10	0.493
4	rate ≤ 0.065	C, D	0.2000	0.693	5, 6	5, 6	0.394
5	warm-up = no	D, C	0.1571	0.840	1, 3, 7, 11	**none**	0.287
6	rate ≤ 0.065	C, D	0.2966	0.432	5, 6	none	0.262

(Deck: five rounds; notes: six.) **A question can come back** (rounds 1, 4, 6) — the weights changed, so it is a different problem. The committee's mistake count can go **up** (round 4), but the exponential loss falls **every** round. After round 5 all twelve are right.

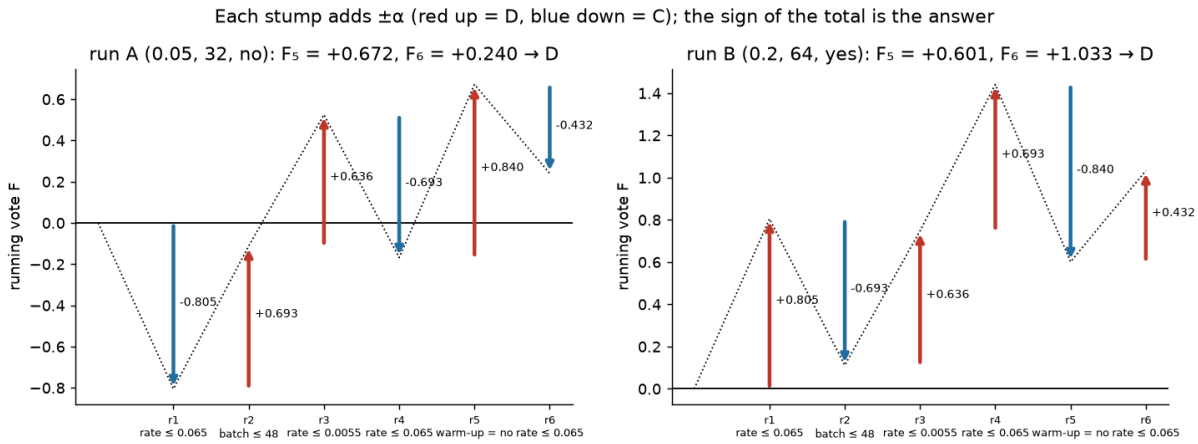


Notes Fig. 4 — The six stumps of the committee.

The vote. Each stump says ±1; multiply by its α; add; the sign is the answer. Contributions $\alpha_t h_t(x)$ (5 rounds):

run	out	r1 rate	r2 batch	r3 rate	r4 rate	r5 warm	F_5	says
1	D	+0.805	−0.693	+0.636	+0.693	−0.840	+0.601	D
2	D	+0.805	+0.693	+0.636	+0.693	+0.840	+3.667	D
5, 6	D	−0.805	+0.693	+0.636	−0.693	+0.840	+0.672	D
7	C	−0.805	−0.693	+0.636	−0.693	+0.840	−0.715	C
10	C	−0.805	+0.693	+0.636	−0.693	−0.840	−1.008	C
12	C	−0.805	−0.693	−0.636	−0.693	−0.840	−3.667	C
A	?	−0.805	+0.693	+0.636	−0.693	+0.840	**+0.672**	**D**
B	?	+0.805	−0.693	+0.636	+0.693	−0.840	**+0.601**	**D**

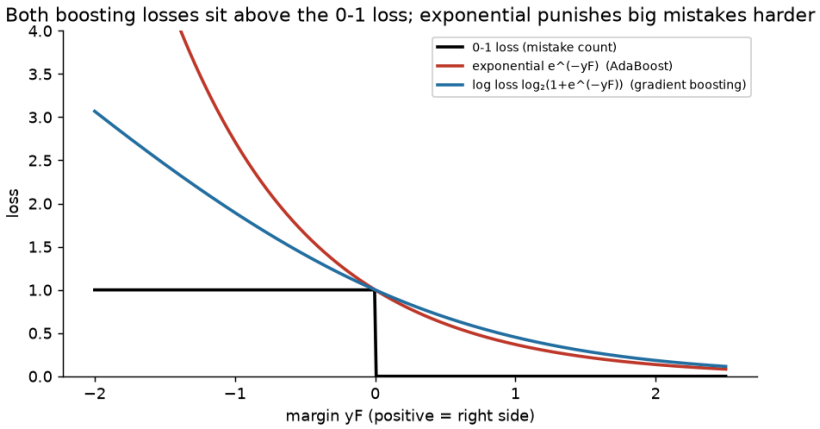
(Full twelve-run table on slide 77.) Both A and B are D **by a small margin**: the stumps disagree. After round 6 (+0.432 for B, −0.432 for A): **B = 1.033**, **A = 0.240** — A is still D, only just. A looks like runs 5 and 6 (small rate, small batch, no warm-up): exactly where a single stump fails and the committee is unsure.



One arrow per stump. AdaBoost gives a **class and a margin**, not a probability.

A8 Same recipe, different loss: AdaBoost = boosting on e^(−yF)

Step 1 — The loss. $L = e^{-yF}$ with $y = \pm 1$. Right with a big margin ($yF \gg 0$): tiny. Wrong ($yF < 0$): large, and growing exponentially.



Both boosting losses lie above the 0-1 loss (so they bound the error count). The exponential punishes big mistakes harder than log loss — which also makes AdaBoost sensitive to mislabelled points.

Step 2 — The "residual". $dL/dF = -y\,e^{-yF}$. So the number playing the residual's role for run i has size $e^{-y_i F_i}$ — **the run's weight, before normalising.**

Step 3 — Check after two rounds ($F_2 = \alpha_1 h_1 + \alpha_2 h_2$):

run	1	2	3	4	5	6	7	8	9
F_2	+0.11	+1.50	+0.11	+0.11	−0.11	−0.11	−1.50	−1.50	−1.50
e^{-yF_2}	0.894	0.224	0.894	0.894	1.118	1.118	0.224	0.224	0.224
÷ 7.16	0.1250	0.0312	0.1250	0.1250	0.1562	0.1562	0.0312	0.0312	0.0312
update rule	0.1250	0.0312	0.1250	0.1250	0.1563	0.1563	0.0312	0.0312	0.0312

The last two rows agree: **AdaBoost weights are the exponential-loss residuals, rescaled to add to 1.** A mistake ($yF < 0$) has $e^{-yF} > 1$ ⇒ large weight.

Step 4 — Derive the whole algorithm from the loss (notes boxes D1–D6). Let $w_i = e^{-y_i F_{\text{old}}(x_i)}$.

1. $E_{\text{new}} = \sum_i e^{-y_i(F_{\text{old}}+\alpha h)} = \sum_i w_i e^{-\alpha y_i h(x_i)}$.
2. Split into right and wrong: $E_{\text{new}} = W_{\text{right}} e^{-\alpha} + W_{\text{wrong}} e^{\alpha}$.
3. **Which stump?** For any α > 0, $e^{\alpha} > e^{-\alpha}$, so with the total fixed, E_{new} is smallest when W_{wrong} is smallest ⇒ **the smallest weighted error**.
4. **Which α?** $dE/d\alpha = 0 \Rightarrow e^{2\alpha} = W_{\text{right}}/W_{\text{wrong}} = (1-\varepsilon)/\varepsilon$ ⇒ the formula.
5. **New weights:** $e^{-y_i F_{\text{new}}} = w_i e^{-\alpha y_i h(x_i)}$ — exactly the update rule; ÷Z normalises. **A run's weight = its share of the total exponential loss.**

A9 Why the committee gets better: the training-error bound

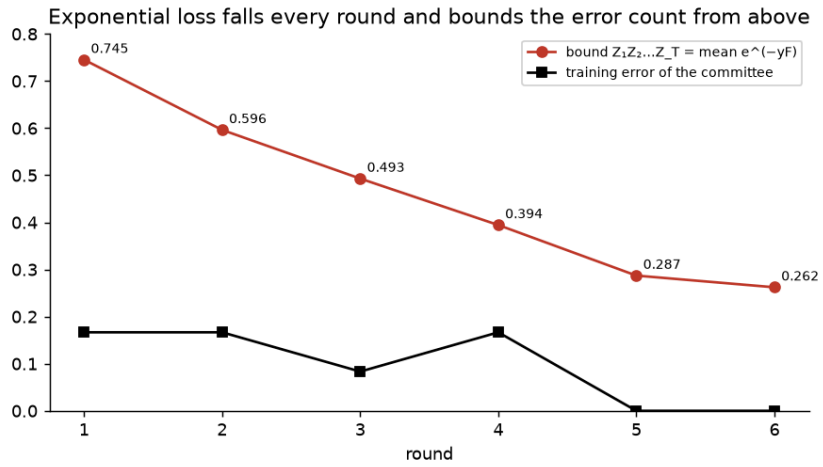
Step 1 — Mistake ⇒ large loss. If $y_i F \le 0$ then $e^{-y_i F} \ge 1$; if right, $e^{-y_i F} > 0$. Either way $1[\text{wrong}] \le e^{-y_i F}$.

Step 2 — Average. $\dfrac{\#\text{wrong}}{N} \le \dfrac{1}{N}\sum_i e^{-y_i F_T(x_i)}$.

Step 3 — Each round multiplies the total loss by Z_t (starting from N at $F = 0$), so

$$\text{training error} \le Z_1 Z_2 \cdots Z_T, \qquad Z_t = 2\sqrt{\varepsilon_t(1-\varepsilon_t)} < 1 \text{ whenever } \varepsilon_t \ne \tfrac{1}{2}.$$

Step 4 — Numbers. $Z_1 = 0.7454$, $Z_2 = 0.8000$, $Z_3 = 0.8268$; product **0.4930** = the mean exponential loss after round 3 computed directly ✓. Every round with ε < ½ shrinks the bound. And ε ≤ ½ is guaranteed when each leaf says its weighted majority (each leaf is wrong on at most half its weight) ⇒ α ≥ 0.



The bound falls monotonically; the error count can bounce (round 4) but always stays under it.

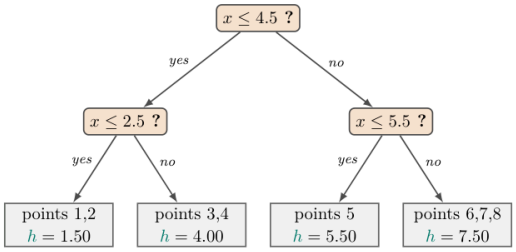
A10 AdaBoost.R2 for regression

Step 1 — What changes. The weak learner is a **two-level regression tree** (up to 4 leaves) whose leaves predict the **weighted average** of y; the error is measured **relative to the worst mistake**; the final answer is a **weighted median**.

Step 2 — Round 1: grow the tree (all weights 1/8). First question by smallest weighted squared error:

question	groups	predictions	weighted SSE
x ≤ 1.5	1 / 7	1.50 / 5.36	4.232
x ≤ 2.5	2 / 6	1.50 / 6.00	2.062
x ≤ 3.5	3 / 5	2.33 / 6.40	1.983
x ≤ 4.5	4 / 4	2.75 / 7.00	**1.344**
x ≤ 5.5	5 / 3	3.30 / 7.50	1.725
x ≤ 6.5	6 / 2	3.92 / 7.75	3.104
x ≤ 7.5	7 / 1	4.36 / 8.50	3.982

Then left by x ≤ 2.5, right by x ≤ 5.5 ⇒ leaves {1, 2}: 1.50; {3, 4}: 4.00; {5}: 5.50; {6, 7, 8}: 7.50.



Notes Fig. 2 — Each leaf predicts the (weighted) average of its points. Your turn 1: $(7 + 7 + 8.5)/3 = 7.50$.

Step 3 — Errors relative to the worst. $\delta_i = |y_i - h(x_i)|$, $\Delta = \max_i \delta_i$, $\ell_i = \delta_i/\Delta \in [0, 1]$.

i	1–5	6	7	8
h	exact	7.50	7.50	7.50
δ	0	0.5	0.5	1.0
ℓ	0	0.5	0.5	1.0

$\Delta = 1$. (Other choices: ℓ^2 or $1 - e^{-\ell}$; the method is the same.)

Step 4 — Average loss, shrink factor, confidence.

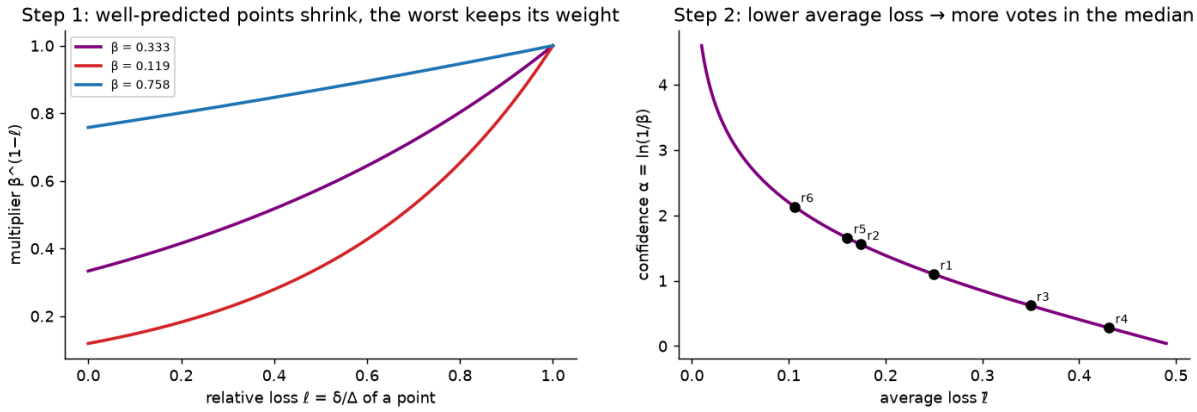
$$\bar{\ell} = \sum w_i \ell_i = \tfrac{1}{8}(0.5 + 0.5 + 1) = 0.25, \quad \beta = \frac{\bar{\ell}}{1 - \bar{\ell}} = \frac{1}{3}, \quad \alpha = \ln \frac{1}{\beta} = \ln 3 = 1.0986.$$

$\beta < 1 \iff \bar{\ell} < \tfrac{1}{2}$. If $\bar{\ell} \ge \tfrac{1}{2}$ the tree is no better than a coin toss on average ⇒ $\beta \ge 1$, $\alpha \le 0$ ⇒ **stop**.

Step 5 — Re-weight: $w_i \leftarrow w_i \beta^{1-\ell_i}/Z$. Perfect ($\ell = 0$): × β (shrinks). Worst ($\ell = 1$): × 1 (stays). In between: × $\sqrt{\beta} = 0.577$ for $\ell = 0.5$.

i	ℓ	multiplier	w × mult.	w after
1–5	0	0.3333	0.0417	**0.0872**
6, 7	0.5	0.5774	0.0722	**0.1511**
8	1	1.0000	0.1250	**0.2617**

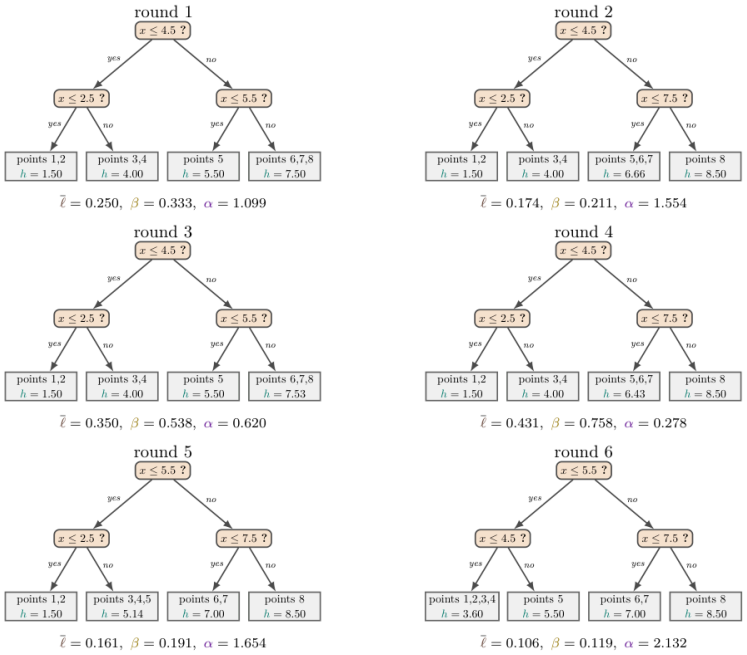
$Z_1 = 5(0.0417) + 2(0.0722) + 0.125 = 0.4777$. Point 8 more than doubled; the next tree tries harder on 6, 7, 8. (Shrinking the good ones vs growing the bad ones is the same thing after ÷Z: only ratios matter.)



Left: how much each point's weight is multiplied, against its loss, for three values of β. Right: a learner's confidence against its average loss, with the six rounds marked.

Step 6 — Rounds 1–6:

round	Δ	$\bar{\ell}$	β	α = ln(1/β)	Z
1	1.000	0.2500	0.3333	1.0986	0.4777
2	1.164	0.1745	0.2113	1.5544	0.3163
3	0.966	0.3497	0.5377	0.6204	0.6881
4	0.927	0.4310	0.7576	0.2776	0.8593
5	1.136	0.1606	0.1913	1.6537	0.2898
6	2.098	0.1061	0.1186	2.1317	0.1819



Notes Fig. 3 — The trees differ: changing weights change both the best question and each leaf's weighted average.

Step 7 — Special case = classification AdaBoost. If every point is perfect ($\ell = 0$) or worst ($\ell = 1$), $\bar{\ell} = \varepsilon$, $\beta = \varepsilon/(1-\varepsilon)$; right × β, wrong × 1 ⇒ wrong:right ratio $1/\beta = (1-\varepsilon)/\varepsilon = e^{2\alpha_{\text{class}}}$ — the same weights. And $\ln(1/\beta) = 2\alpha_{\text{class}}$: doubling every confidence changes neither a weighted median nor the sign of a weighted vote.

A11 The weighted median

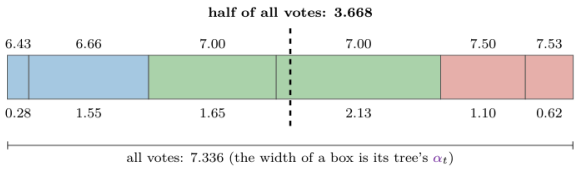
Step 1 — Ordinary median. Trees say 6.4, 7.0, 7.5 ⇒ middle one 7.0.

Step 2 — Weighted median. Each tree carries α_t votes. Sort predictions; keep a **running total** of votes; the answer is the prediction where the running total **first reaches half** of all votes. Simplest case at $x = 6$ after two trees: tree 1 says 7.50 ($\alpha = 1.099$), tree 2 says 6.66 ($\alpha = 1.554$); half of 2.653 is 1.327; tree 2 alone already has 1.554 ⇒ **6.66**. The more confident tree wins.

Step 3 — Six trees at $x = 6$ (total 7.336, half 3.668):

tree	prediction	α	running total
4	6.43	0.278	0.278
2	6.66	1.554	1.832
5	7.00	1.654	3.486
6	**7.00**	2.132	**5.617 ← median**
1	7.50	1.099	6.716
3	7.53	0.620	7.336

⇒ **7.00** (truth 7).

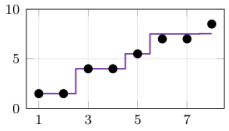


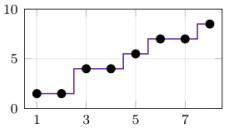
Notes Fig. 5 — Box width = votes. Wherever the halfway line falls, that box's prediction is the answer.

Step 4 — Why the median minimises weighted absolute distance. $S(m) = \sum_t \alpha_t |m - v_t|$. Between two neighbouring values, $dS/dm = \sum_{v_t<m} \alpha_t - \sum_{v_t>m} \alpha_t$ (left confidence – right confidence). Moving right helps while the left total is smaller; it stops helping where they balance — the weighted median. *Why not an average?* A single confident but wild tree can drag an average far; a median barely moves.

Combination rules side by side:

	every tree gives	final answer
gradient boosting (L12c)	a correction νγ	sum of corrections added to the start
AdaBoost, classification	a vote ±1	sign of $\sum \alpha_k h_k$
AdaBoost.R2	a number $h_t(x)$	weighted median, weights α_t





Notes Fig. 6 — On these noise-free points the ensemble ends up matching every point. On noisy data it shouldn't — choose the number of rounds on validation data.

A12 The recipes and common confusions

AdaBoost (classification) in five lines.

1. $w_i = 1/N$.
2. Pick the stump with the smallest weighted error ε (each leaf = its weighted majority).
3. $\alpha = \frac{1}{2} \ln \frac{1-\varepsilon}{\varepsilon}$.
4. $w_i \leftarrow w_i e^{-\alpha y_i h(x_i)}$, then divide by the total.
5. Repeat; predict $\operatorname{sign} \sum_k \alpha_k h_k(x)$.

AdaBoost.R2 in six lines. (1) $w_i = 1/N$. (2) Grow a small tree on the weights (leaves = weighted means). (3) $\delta_i = |y_i - h(x_i)|$, $\ell_i = \delta_i / \Delta$, $\bar{\ell} = \sum w_i \ell_i$ (stop if ≥ ½). (4) $\beta = \bar{\ell}/(1-\bar{\ell})$, $\alpha = \ln(1/\beta)$. (5) $w_i \leftarrow w_i \beta^{1-\ell_i}$, normalise. (6) Repeat; predict the weighted median with weights α_t.

Common confusions.

- *Why not use the best stump alone?* It is wrong on runs 5, 6 and one question cannot separate them. Adding stumps builds a more complex boundary from simple pieces.
- *Why do wrong runs get heavier?* The next stump minimises the **weighted** error; heavy mistakes are expensive, so it is steered towards them.
- *Can ε exceed ½?* Not when each leaf predicts its weighted majority.
- *Different from bagging?* Bagging grows independent trees and averages; AdaBoost grows them in sequence, each shaped by the previous mistakes, and the vote is weighted.
- *Where do the other two come from?* Gradient boosting and XGBoost keep the idea but pass mistakes as residuals (or slope + curvature). AdaBoost **is** gradient boosting for the exponential loss (A8).
- *Why relative error in R2?* It puts every round on a 0–1 scale whatever the units of y. Price: Δ is set by one point, so an outlier can dominate the scale.
- *R2 vs gradient boosting for regression?* GB fits residuals and **adds** (small steps); R2 fits the original y with changed weights and takes a **weighted median**.

Part B Slide by Slide (slides 65–81)

Slide 65 Part 4: AdaBoost (divider)

Colours: run weight, mistake, and each stump's numbers by its question.

Slide 66 AdaBoost: the procedure

Shown in A1.

Slide 67 AdaBoost: definitions

AdaBoost: definitions

- **Labels.** $y_i = +1$ if the run diverged (**D**), -1 if it converged (**C**). A stump gives $h(x) = \pm 1$: each leaf predicts the class with the larger total weight in that leaf.
- **Run weights** $w_i \geq 0$ with $\sum_i w_i = 1$; at the start $w_i = \frac{1}{12}$.
- **Weighted error** of a stump: $\varepsilon = \sum_{i:\, h(x_i) \neq y_i} w_i$ (the total weight of its mistakes).
- **Vote weight** $\alpha = \frac{1}{2} \ln \frac{1-\varepsilon}{\varepsilon}$: large when ε is small, 0 when $\varepsilon = \frac{1}{2}$, negative when $\varepsilon > \frac{1}{2}$.
- **Weight update.** $w_i \leftarrow \frac{w_i\, e^{-\alpha\, y_i\, h(x_i)}}{Z}$, so a correct run is multiplied by $e^{-\alpha}$ and a mistake by $e^{+\alpha}$; Z makes the weights add to 1.
- **Model after** T **rounds.** $F(x) = \sum_{t=1}^{T} \alpha_t h_t(x)$; predict **D** if $F(x) > 0$, **C** otherwise.

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Slide 67 — All the symbols in one place (A1).

Slide 68

Why this vote weight

Why this vote weight

Multiplying correct runs by $e^{-\alpha}$ and mistakes by $e^{+\alpha}$ gives total weight

$$Z(\alpha) = (1-\varepsilon)e^{-\alpha} + \varepsilon e^{\alpha}.$$

Choose α to make Z as small as possible: $\frac{dZ}{d\alpha} = -(1-\varepsilon)e^{-\alpha} + \varepsilon e^{\alpha} = 0 \Rightarrow e^{2\alpha} = \frac{1-\varepsilon}{\varepsilon}$, so

$$\boxed{\alpha = \tfrac{1}{2}\ln\frac{1-\varepsilon}{\varepsilon}}, \qquad Z = 2\sqrt{\varepsilon(1-\varepsilon)}.$$

What the update does. After normalising, the mistakes carry total weight $\frac{\varepsilon e^{\alpha}}{Z} = \frac{\varepsilon\sqrt{(1-\varepsilon)/\varepsilon}}{2\sqrt{\varepsilon(1-\varepsilon)}} = \frac{1}{2}$ and the correct runs carry the other $\frac{1}{2}$.

So the stump just used has weighted error exactly $\frac{1}{2}$ on the new weights: it is no better than a random guess, so the next stump must find something different. The loss being reduced is the exponential loss e^{-yF} (a later slide shows this).

Slide 68 — The stump just used is now a coin toss; the loss being reduced is e^{-yF} (A4–A5).

Slides 69–70

Your turn: the vote weight (and answer)

Answer: the vote weight

ε	0.1	0.3	0.5	0.7
α	$\frac{1}{2}\ln 9 = 1.099$	$\frac{1}{2}\ln\frac{7}{3} = 0.424$	$\frac{1}{2}\ln 1 = 0$	$\frac{1}{2}\ln\frac{3}{7} = -0.424$

The stump with $\varepsilon = 0.1$ gets the largest vote. A stump at $\varepsilon = 0.5$ gets no vote at all. A stump with $\varepsilon = 0.7$ is wrong more often than not, so its vote is *negative*: AdaBoost uses it by flipping its answer. (If $\varepsilon \to 0$ then $\alpha \to \infty$: a perfect stump would take over the vote, so implementations cap α.)

Slide 70 — Negative vote = flip the stump; perfect stump ⇒ α → ∞, so cap it.

Slide 71 Round 1: choose the stump

Round 1: choose the stump

All twelve weights are $\frac{1}{12}$, so the weighted error is (number of mistakes)/12. Best five questions:

question	mistakes	ε
rate ≤ 0.065	2	0.1667
rate ≤ 0.02	3	0.2500
rate ≤ 0.0055	4	0.3333
rate ≤ 0.2	4	0.3333
batch ≤ 48	4	0.3333

Winner rate ≤ 0.065. Yes-group (runs 5–12): 6 **C**, 2 **D**, so it predicts **C**. No-group (runs 1–4): all **D**, so it predicts **D**. Mistakes: runs 5 and 6 (diverged, but low rate).

$$\varepsilon_1 = w_5 + w_6 = \tfrac{2}{12} = 0.1667, \qquad \alpha_1 = \tfrac{1}{2}\ln\frac{1-0.1667}{0.1667} = \tfrac{1}{2}\ln 5 = 0.8047$$

Slide 71 — Mistakes: runs 5 and 6 (A2).

Slide 72 Round 1: update the weights

Round 1: update the weights

run	y	$h_1(x)$	correct?	w before	multiplier	$w\times$ mult.	w after
1	D	D	yes	0.0833	0.447	0.0373	0.0500
2	D	D	yes	0.0833	0.447	0.0373	0.0500
3	D	D	yes	0.0833	0.447	0.0373	0.0500
4	D	D	yes	0.0833	0.447	0.0373	0.0500
5	D	C	no	0.0833	2.236	0.1863	0.2500
6	D	C	no	0.0833	2.236	0.1863	0.2500
7	C	C	yes	0.0833	0.447	0.0373	0.0500
8	C	C	yes	0.0833	0.447	0.0373	0.0500
9	C	C	yes	0.0833	0.447	0.0373	0.0500
10	C	C	yes	0.0833	0.447	0.0373	0.0500
11	C	C	yes	0.0833	0.447	0.0373	0.0500
12	C	C	yes	0.0833	0.447	0.0373	0.0500

$e^{-\alpha_1} = 0.447$ for a correct run, $e^{+\alpha_1} = 2.236$ for a mistake. $Z = \sum = 0.7454$; dividing by Z: 0.25 for each mistake, 0.05 for every other run.

Slide 72: AdaBoost round one. The ten correct runs are multiplied by 0.447 and the two mistakes by 2.236, so after normalizing each mistake carries weight 0.25.

Slides 73–74 Your turn: the second stump (and answer)

Answer: the second stump

1. Run 10 (**C**, predicted **D**) and runs 1, 3, 4 (**D**, predicted **C**).

2. $\varepsilon = 4 \times 0.05 =$ 0.200. Ordinary error rate $\frac{4}{12} = 0.333$: the weighted error is smaller because the mistakes are runs that already count little.

3. $\alpha = \frac{1}{2} \ln \frac{0.8}{0.2} = \frac{1}{2} \ln 4 =$ 0.693.

4. The round-1 stump errs on runs 5 and 6: $\varepsilon = 0.25 + 0.25 = 0.5$, so $\alpha = 0$. Reusing it adds nothing, which is the result of the previous slide: after an update, the stump just used has weighted error $\frac{1}{2}$.

Tie. batch ≤ 48, warm-up $=$ no and rate ≤ 0.0055 all have $\varepsilon = 0.2$. Rule used in this deck: prefer a feature not asked before, then the order rate, batch, warm-up. So round 2 asks batch.

Slide 74: the second stump has weighted error 0.2 and earns a vote of 0.693. Reusing the first stump would score exactly 0.5 and earn no vote.

Slide 75 Round 2: update the weights

Round 2: update the weights

run	y	$h_2(x)$	correct?	w before	multiplier	$w \times$ mult.	w after
1	D	C	no	0.0500	2.000	0.1000	0.1250
2	D	D	yes	0.0500	0.500	0.0250	0.0312
3	D	C	no	0.0500	2.000	0.1000	0.1250
4	D	C	no	0.0500	2.000	0.1000	0.1250
5	D	D	yes	0.2500	0.500	0.1250	0.1563
6	D	D	yes	0.2500	0.500	0.1250	0.1563
7	C	C	yes	0.0500	0.500	0.0250	0.0312
8	C	C	yes	0.0500	0.500	0.0250	0.0312
9	C	C	yes	0.0500	0.500	0.0250	0.0312
10	C	D	no	0.0500	2.000	0.1000	0.1250
11	C	C	yes	0.0500	0.500	0.0250	0.0312
12	C	C	yes	0.0500	0.500	0.0250	0.0312

$\alpha_2 = 0.693$: multipliers 0.500 (correct), 2.000 (mistake); $Z = 0.8000$. Runs 1, 3, 4, 10 carry 0.125 each; runs 5, 6 fall to 0.156.

Slide 75: AdaBoost round two. The four mistakes are doubled and the correct runs halved, before normalizing.

Slide 76

Five rounds

Five rounds

round	stump	yes, no predict	ε	α	stump's mistakes	model's mistakes	$\overline{e^{-yF}}$
1	rate ≤ 0.065	**C**, **D**	0.1667	0.805	5, 6	5, 6	0.745
2	batch ≤ 48	**D**, **C**	0.2000	0.693	1, 3, 4, 10	5, 6	0.596
3	rate ≤ 0.0055	**C**, **D**	0.2188	0.636	7, 8, 9, 10	10	0.493
4	rate ≤ 0.065	**C**, **D**	0.2000	0.693	5, 6	5, 6	0.394
5	warm-up = no	**D**, **C**	0.1571	0.840	1, 3, 7, 11	none	0.287

The model's mistakes use all stumps so far ($F = \sum \alpha_t h_t$). The *count* of mistakes can go up (round 4) but the exponential loss $\overline{e^{-yF}}$ (mean over the 12 runs) falls every round. After round 5 the model classifies all twelve runs correctly.

Slide 76 — Mistake count can rise (round 4); exponential loss falls every round; all twelve right after round 5 (A7).

Slide 77

Contribution of each stump to each run's vote

Contribution of each stump to each run's vote

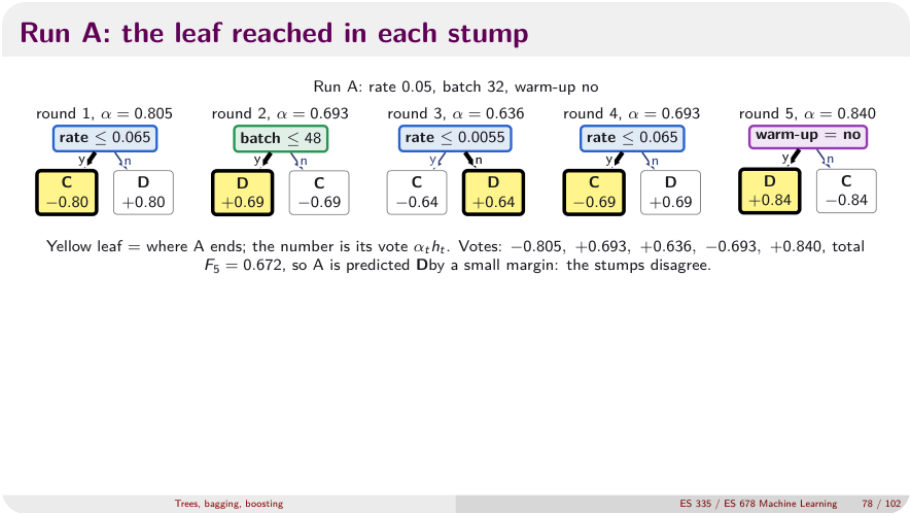
run	out	r1 rate	r2 batch	r3 rate	r4 rate	r5 warm	F_5	predicts
1	**D**	+0.805	−0.693	+0.636	+0.693	−0.840	+0.601	**D**
2	**D**	+0.805	+0.693	+0.636	+0.693	+0.840	+3.667	**D**
3	**D**	+0.805	−0.693	+0.636	+0.693	−0.840	+0.601	**D**
4	**D**	+0.805	−0.693	+0.636	+0.693	+0.840	+2.281	**D**
5	**D**	−0.805	+0.693	+0.636	−0.693	+0.840	+0.672	**D**
6	**D**	−0.805	+0.693	+0.636	−0.693	+0.840	+0.672	**D**
7	**C**	−0.805	−0.693	+0.636	−0.693	+0.840	−0.715	**C**
8	**C**	−0.805	−0.693	+0.636	−0.693	−0.840	−2.394	**C**
9	**C**	−0.805	−0.693	+0.636	−0.693	−0.840	−2.394	**C**
10	**C**	−0.805	+0.693	+0.636	−0.693	−0.840	−1.008	**C**
11	**C**	−0.805	−0.693	−0.636	−0.693	+0.840	−1.988	**C**
12	**C**	−0.805	−0.693	−0.636	−0.693	−0.840	−3.667	**C**
A	?	−0.805	+0.693	+0.636	−0.693	+0.840	+0.672	**D**
B	?	+0.805	−0.693	+0.636	+0.693	−0.840	+0.601	**D**

Entry $= \alpha_t\, h_t(x)$: plus the stump's vote weight if its leaf says **D**, minus if **C**. F_5 is the row sum. All twelve training runs are classified correctly. New runs: A $F_5 = 0.672$, B $F_5 = 0.601$.

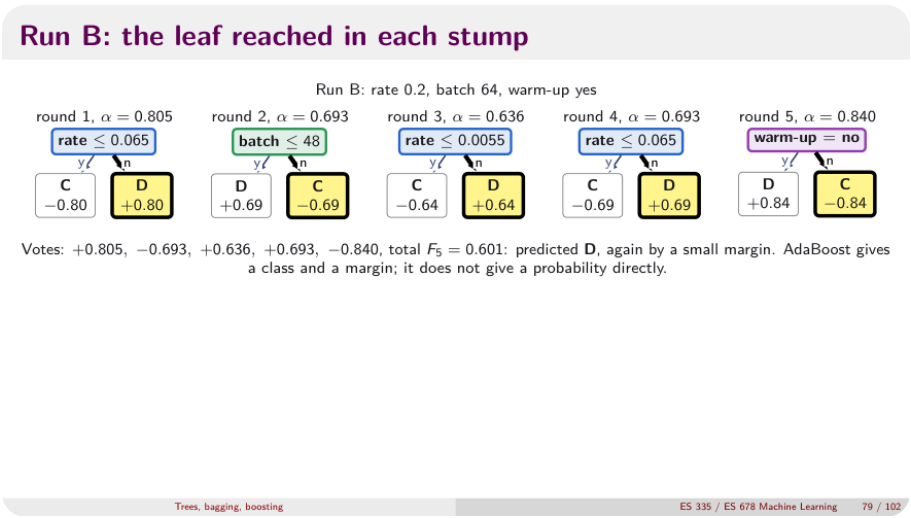
Slide 77 — A: $F_5 = 0.672$; B: $F_5 = 0.601$.

Slides 78–79

Runs A and B: the leaf reached in each stump



Slide 78 — Run A: D by a small margin; the stumps disagree.



Slide 79 — AdaBoost gives a class and a margin, not a probability.

Slide 80 Same recipe, different loss

Same recipe, different loss

Part 3 reduced the **log loss** step by step. AdaBoost reduces the **exponential loss** $L = e^{-yF}$ for $y = \pm 1$ in exactly the same stagewise way. The derivative with respect to the score is $\frac{dL}{dF} = -y\, e^{-yF}$, so the number that plays the role of the residual for run i is

$$\text{(its size) } e^{-y_i F_i} \;=\; \text{the run's weight, before dividing by the total.}$$

Check after two rounds, $F = \alpha_1 h_1 + \alpha_2 h_2$:

run	1	2	3	4	5	6	7	8	9	10	11	12
F_2	+0.11	+1.50	+0.11	+0.11	−0.11	−0.11	−1.50	−1.50	−1.50	−0.11	−1.50	−1.50
e^{-yF_2}	0.894	0.224	0.894	0.894	1.118	1.118	0.224	0.224	0.224	0.894	0.224	0.224
÷7.16	0.1250	0.0312	0.1250	0.1250	0.1562	0.1562	0.0312	0.0312	0.0312	0.1250	0.0312	0.0312
update	0.1250	0.0312	0.1250	0.1250	0.1563	0.1563	0.0312	0.0312	0.0312	0.1250	0.0312	0.0312

The last two rows agree: the AdaBoost weights are the exponential-loss residuals, rescaled to add to 1. A mistake ($yF < 0$) makes $e^{-yF} > 1$ and so a large weight.

Slide 80 — Weights = exponential-loss residuals, rescaled (A8).

Slide 81 AdaBoost: summary of the procedure

AdaBoost: summary of the procedure

step	what you do	which number is used
start	$w_i = \frac{1}{12}$, labels $y_i = \pm 1$	none
(a) stump	smallest $\varepsilon = \sum_{\text{mistakes}} w_i$; each leaf predicts its weighted majority	the weights of the mistakes
(b) vote weight	$\alpha = \frac{1}{2} \ln \frac{1-\varepsilon}{\varepsilon}$	ε of this stump
(c) new weights	$w_i \leftarrow w_i e^{-\alpha y_i h(x_i)} / Z$	α of this stump
predict	$F(x) = \sum_t \alpha_t h_t(x)$; **D** if $F > 0$	one signed vote per stump (the table of contributions)
stop	when held-out error stops falling	–

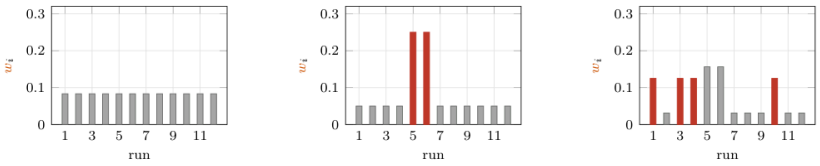
Slide 81 — Stop when held-out error stops falling.

Part B′ — The Notes (L12e and L12f)

Notes L12e (AdaBoost classification), §1–2 — round 1 step by step

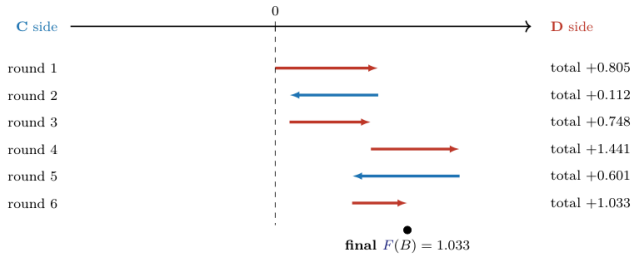
Six steps: equal weights; weighted error of every stump; keep the smallest; α; re-weight; "what just happened" (A2–A5).

Notes L12e, §3 — rounds 2 and 3



Notes Fig. 3 — Red = runs the previous stump got wrong.

Notes L12e, §4 — the final answer: a weighted vote



Notes Fig. 5 — Rounds 2 and 5 vote C, but the total ends clearly on the D side. Run A's six votes add to only 0.240. Committee mistakes after rounds 1–6: 2, 2, 1, 2, 0, 0.

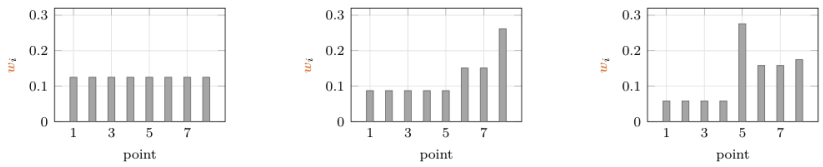
Notes L12e, §5 — derive-it-yourself boxes

D1: $E_{\text{new}} = \sum w_i e^{-\alpha y_i h}$. D2: $= W_{\text{right}} e^{-\alpha} + W_{\text{wrong}} e^{\alpha}$. D3: choose the smallest W_{wrong}. D4: $\alpha = \frac{1}{2} \ln \frac{1-\varepsilon}{\varepsilon}$. D5: $Z = 2\sqrt{\varepsilon(1-\varepsilon)}$ (0.7454 at ε = 1/6). D6: new loss per run = $w_i e^{-\alpha y_i h}$ = the update. D7: training error ≤ $Z_1 \cdots Z_T$ (A8–A9).

Notes L12e, §7 — check yourself (answers)

1 1/12 · 2 weight, wrong · 3 1/6, 5, 0.805 · 4 0, no say · 5 $e^{-\alpha} = 0.447$, $e^{\alpha} = 2.236$ · 6 0.250, 0.50 · 7 0.5, different · 8 α_k, weighted · 9 larger · 10 exponential.

Notes L12f (AdaBoost.R2), §1–3 — problem, round 1, rounds 2–6



Notes Fig. 4 — Weights follow the worst-predicted points. Your turns 1–3: A10.

Notes L12f, §4 — the final prediction

Weighted median (A11). Your turn 4 at $x = 5$: sorted 5.14 (α 1.654, running 1.654), 5.50 (0.620, 2.274), 5.50 (1.099, 3.373), **5.50 (2.132, 5.504)**, 6.43, 6.66 ⇒ median **5.50** (truth 5.5). The ensemble's prediction (weighted median) after 1, 2, 3 and 6 trees:

trees	x=1	2	3	4	5	6	7	8
1	1.50	1.50	4.00	4.00	5.50	7.50	7.50	7.50
2	1.50	1.50	4.00	4.00	6.66	6.66	6.66	8.50
3	1.50	1.50	4.00	4.00	5.50	7.50	7.50	7.53
6	1.50	1.50	4.00	4.00	5.50	7.00	7.00	8.50
truth	1.50	1.50	4.00	4.00	5.50	7.00	7.00	8.50

(rows = ensemble after that many trees)

Notes L12f, §5 — derive-it-yourself boxes

D1–D2: the $\ell \in \{0, 1\}$ special case reproduces classification AdaBoost, with $\ln(1/\beta) = 2\alpha_{\text{class}}$. D3: weighted median minimises $\sum \alpha_t |m - v_t|$. D4: β < 1 $\iff \bar{\ell} < \frac{1}{2}$; otherwise stop. D5: β = 1/3: multipliers 0.333, 0.577, 1 for ℓ = 0, 0.5, 1.

Notes L12f, §7 — check yourself (answers)

1 1/8, 1 · 2 weighted average, more · 3 $|y_i - h(x_i)|$, Δ · 4 1, 0.25 · 5 $1 - \bar{\ell}$, 0.333, 1.099 · 6 β, 1, more · 7 0.262, 0.087 · 8 weighted median, α_t, 7.00 · 9 classification, twice · 10 ½, 1.

Part C Practice

1. A stump has weighted error 0.2. Its vote weight? After the update, what is that same stump's weighted error?
2. Compute Z for ε = 0.2 and check it equals $(1-\varepsilon)e^{-\alpha}+\varepsilon e^{\alpha}$ with your α.
3. In round 1, why is the vote weight of "batch ≤ 48" (ε = 1/3) smaller than that of "rate ≤ 0.065"? Compute it.
4. Starting from the round-2 weights (0.125, 0.031, 0.125, 0.125, 0.156, 0.156, 0.031 ×5 for 7–9, 11, 12, and 0.125 for 10), compute ε for rate ≤ 0.0055 (yes says C, no says D) and confirm 0.2188.
5. Why are AdaBoost run weights called exponential-loss residuals?
6. Prove $1[\text{wrong}] \le e^{-yF}$ and use it to bound the training error after three rounds.
7. AdaBoost.R2: with β = 1/3 and $Z_1 = 0.4777$, compute the new weights of points 1 and 8 (each started at 0.125).
8. AdaBoost.R2: a tree has $\bar{\ell} = 0.55$. What happens and why?
9. Five trees predict 3, 4, 4, 6, 9 with confidences 1, 1, 0.5, 2, 0.5. Weighted median? Ordinary mean?
0. Why is AdaBoost more sensitive to a mislabelled training point than gradient boosting with log loss?

1. $\alpha = \frac{1}{2}\ln 4 = 0.693$. Afterwards exactly ½ ⇒ it would get vote 0.
2. $Z = 2\sqrt{0.2 \times 0.8} = 0.8$. Check: $0.8 \times 0.5 + 0.2 \times 2 = 0.4 + 0.4 = 0.8$ ✓.
3. Larger error ⇒ quieter voice: $\frac{1}{2}\ln\frac{2/3}{1/3} = \frac{1}{2}\ln 2 = 0.347$ vs 0.805.
4. rate ≤ 0.0055 yes = runs 11, 12 (C) ⇒ says C; no = the other ten: D weight $0.125 \times 3 + 0.031 + 0.156 \times 2 = 0.719$, C weight $0.031 \times 3 + 0.125 = 0.219$ ⇒ says D. Mistakes: runs 7, 8, 9, 10 ⇒ $3(0.03125) + 0.125 = 0.21875$ ✓.
5. $w_i \propto e^{-y_i F_i} = |dL/dF|$ for $L = e^{-yF}$: each weight is the size of that run's gradient (its share of the loss).
6. If wrong, $yF \le 0 \Rightarrow e^{-yF} \ge 1$; if right, $e^{-yF} > 0$. So error ≤ mean $e^{-yF} = Z_1 Z_2 Z_3 = 0.7454 \times 0.8 \times 0.8268 = 0.493$.
7. Point 1 (ℓ = 0): $0.125 \times \frac{1}{3}/0.4777 = 0.087$. Point 8 ($\ell$ = 1): $0.125/0.4777 = 0.262$.
8. $\beta = 0.55/0.45 = 1.22 > 1$, $\alpha < 0$: on average the tree is no better than a coin toss; adding it with positive voice would not help ⇒ the algorithm stops.
9. Total 5, half 2.5. Running: 3 (1), 4 (2), 4 (2.5) ⇒ median **4**. Mean of the predictions: 5.2 (dragged up by the 9); weighted mean $= (3+4+2+12+4.5)/5 = 5.1$.

10. A mislabelled point keeps $yF < 0$; its exponential loss (and weight) grows like $e^{|F|}$, so it can dominate later rounds. Log loss grows only linearly for large mistakes, so its pull (residual ≤ 1) is bounded.

Next: L12e — XGBoost & the comparison: gradient boosting with curvature in the split score, and penalties λ and γ.